McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3
Mathematical Biology

Instructor: David Earn

Lecture 8
Space
Tuesday 29 October 2024

Announcments

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- **Midterm test:**
  - *Date:* Tuesday 12 November 2024
  - *Time:* 2:30pm–4:30pm
  - *Location:* in class, HH-102

  - Test structure will be discussed in class next week.

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- **Assignment 4** is due the day before the midterm.

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  - Make sure *you personally* can do the question on calculating $\mathcal{R}_0$ on this assignment *before* the midterm test.

Spatial Epidemic Dynamics

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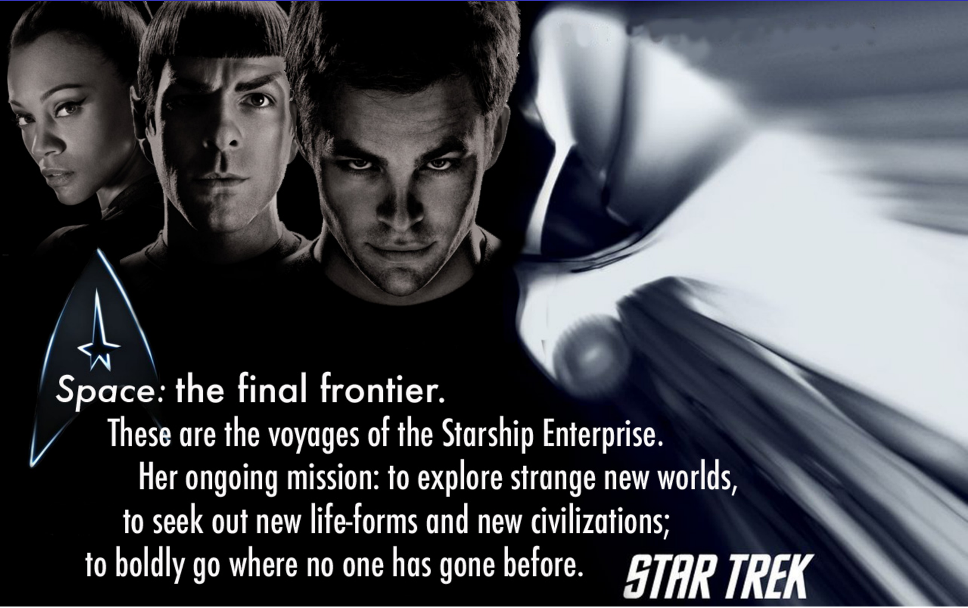

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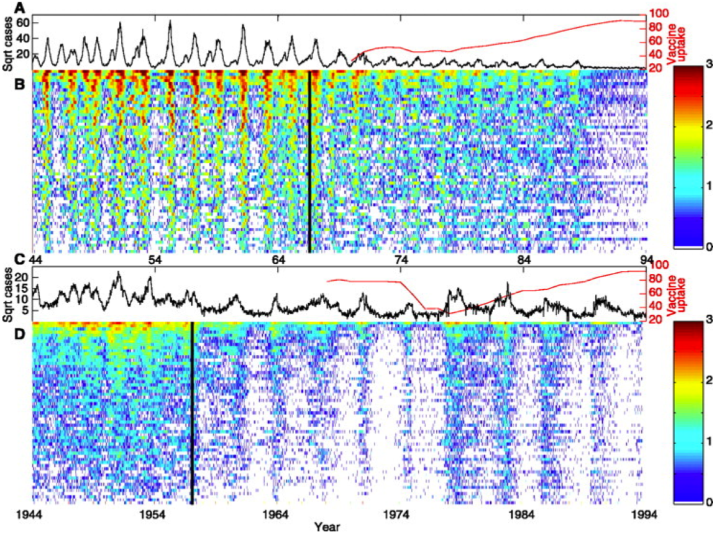
- Can spatial epidemic data suggest improved strategies for control?
- Can we reduce the eradication threshold below $p_{\rm crit} = 1 - \frac{1}{\mathcal{R}_0}$?

Measles and Whooping Cough in 60 UK cities

Measles

Whooping Cough



Rohani, Earn & Grenfell (1999) *Science* **286**, 968–971

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- Can we manipulate epidemics predictably so as to increase probability of eradication?
- Can we eradicate measles?

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- But analytical theory of synchrony in a periodically forced system of differential equations is mathematically demanding...
- So let's consider a much simpler biological model...

The Logistic Map

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- What kinds of dynamics are possible for the Logistic Map?

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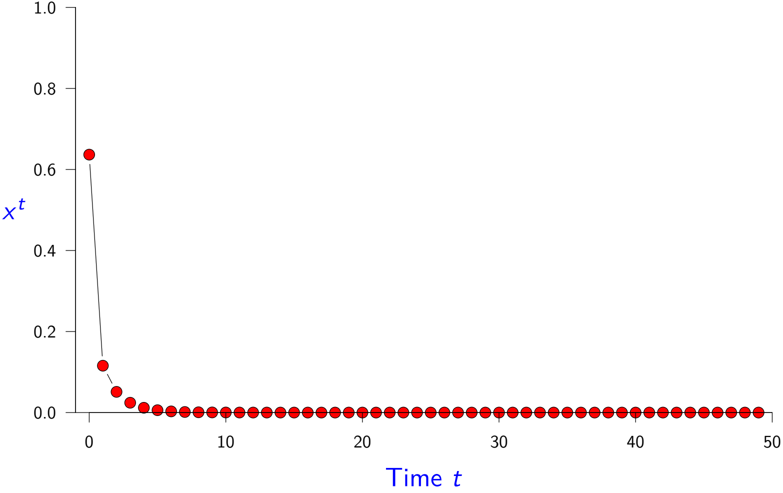
$$x^{t+1} = rx^t(1 - x^t), \quad r = 0.5, \quad x_0 = 0.63662$$



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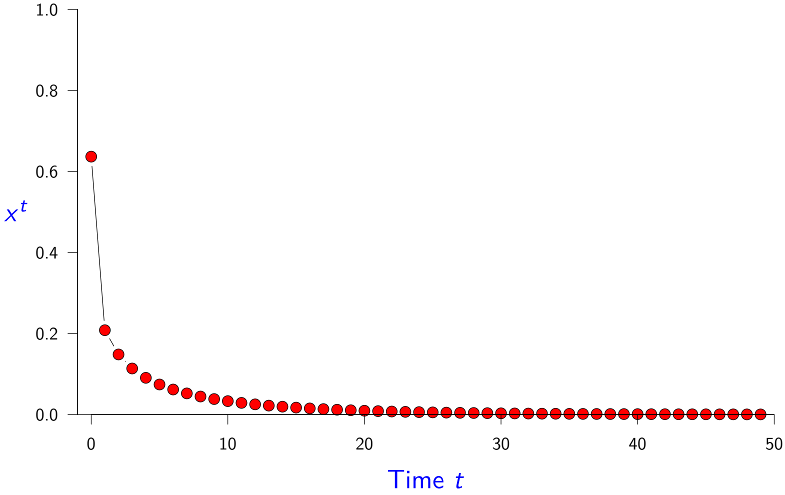

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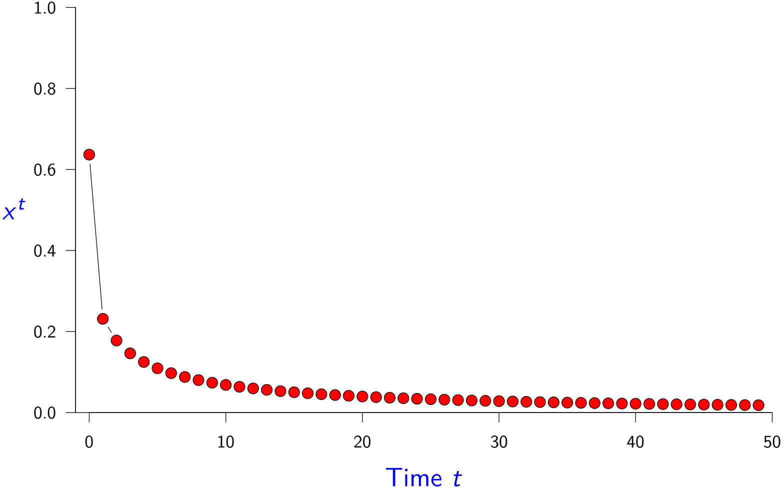

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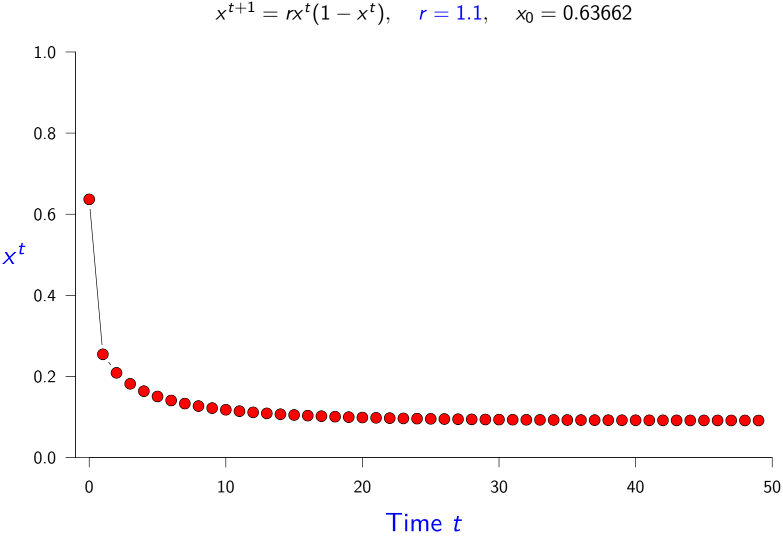

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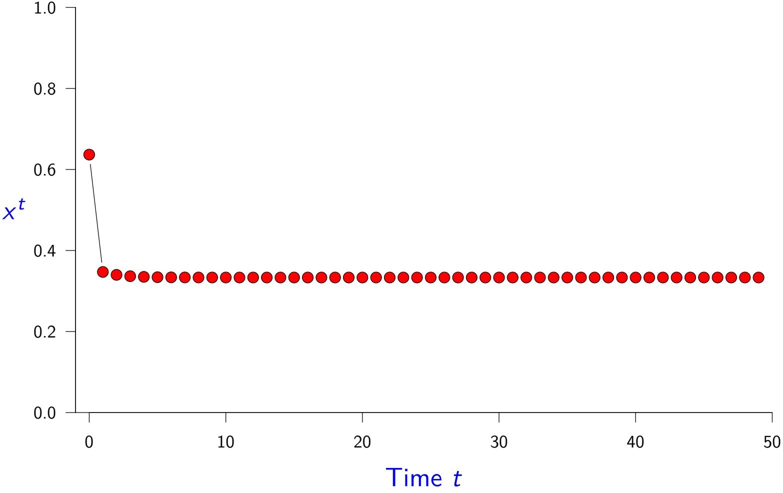

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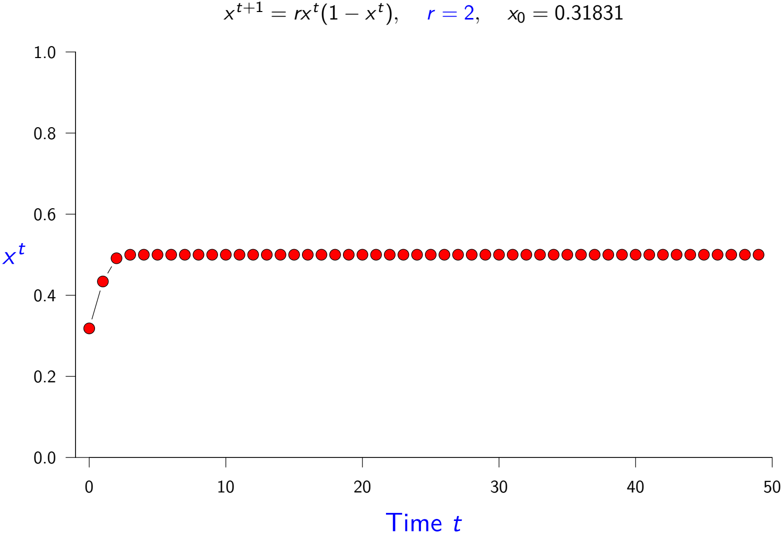

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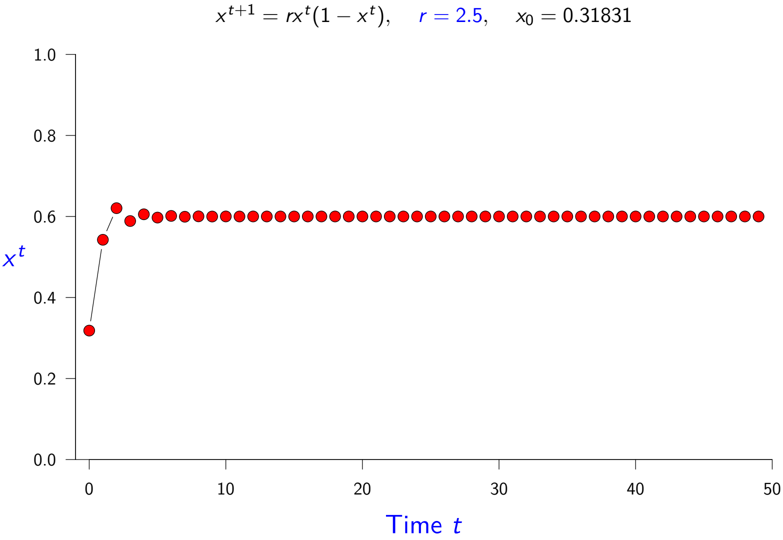

Logistic Map Time Series, $r = 3$

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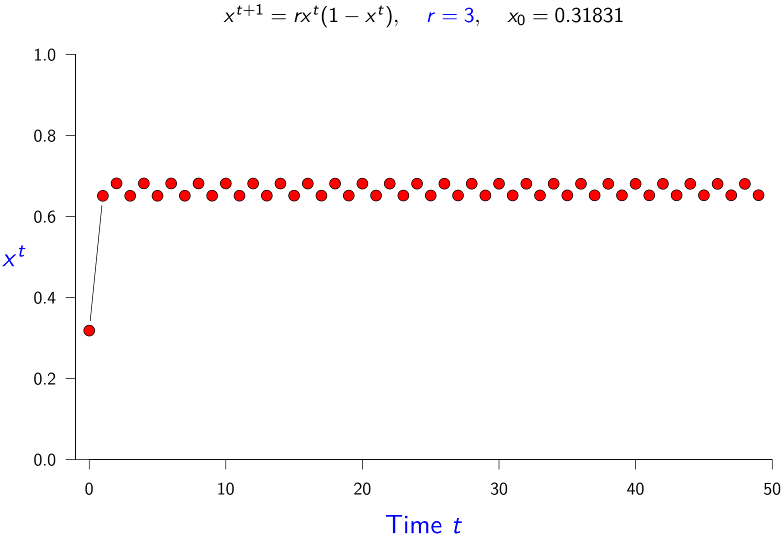

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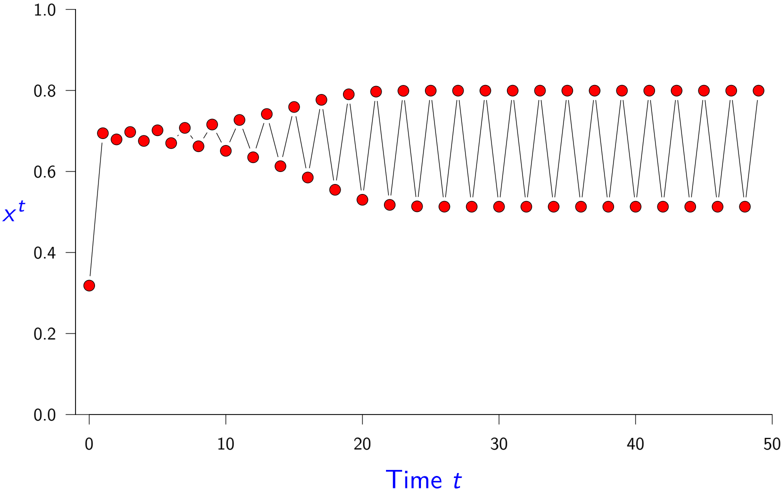

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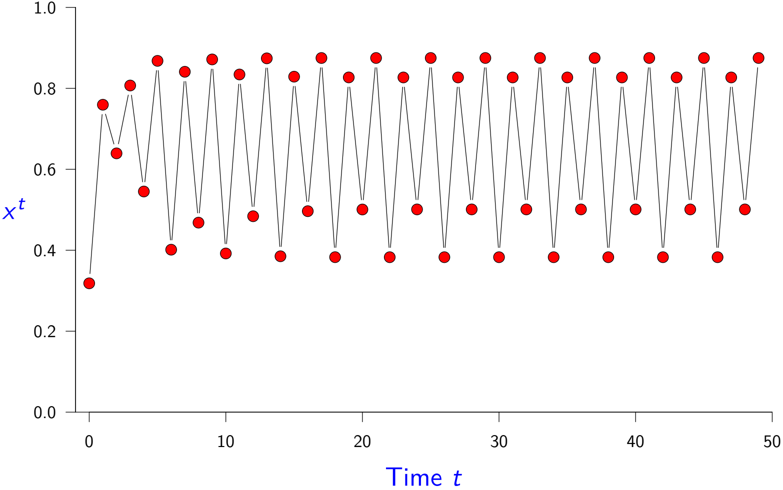

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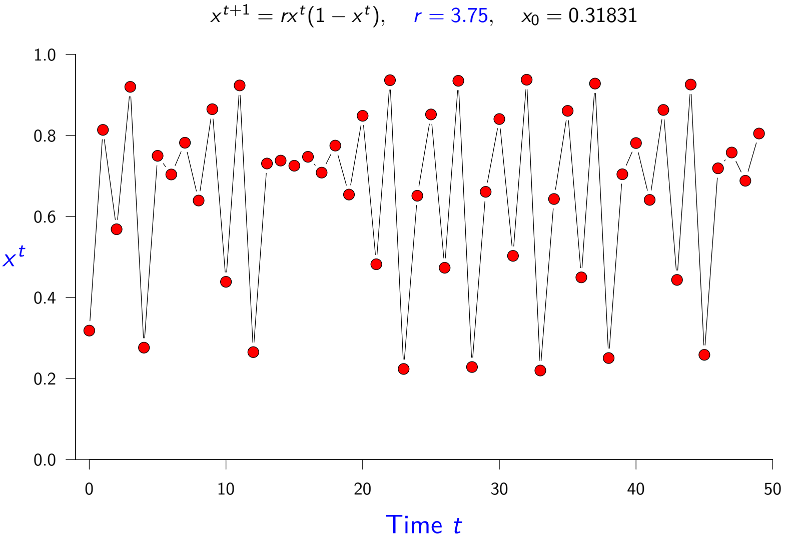

Logistic Map Time Series, $r = 3.83$

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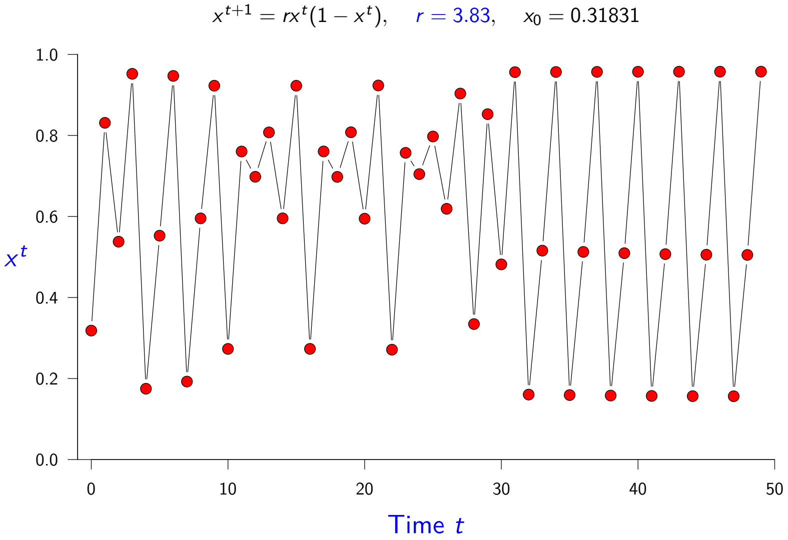

Logistic Map Time Series, $r = 4$

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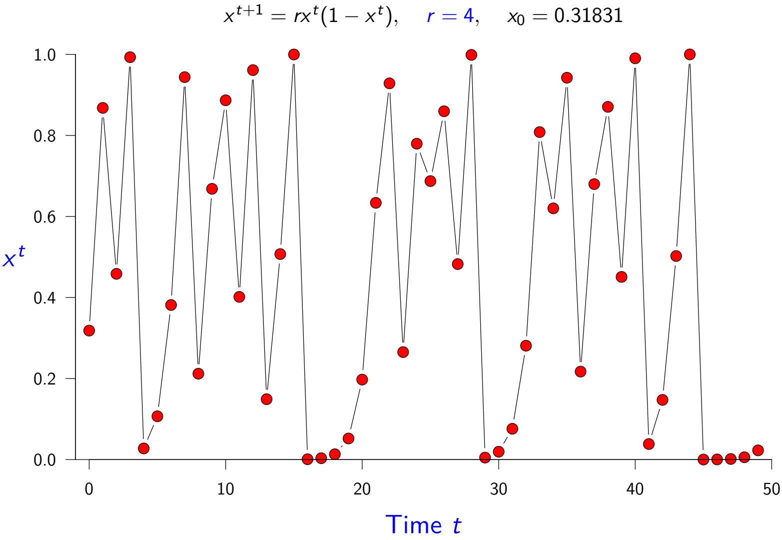

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  - Ignore transient behaviour: just show attractor.

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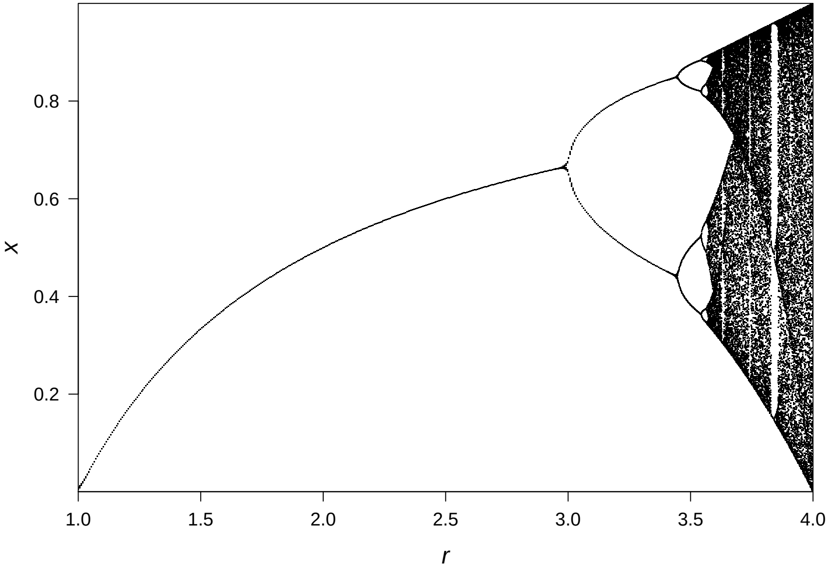
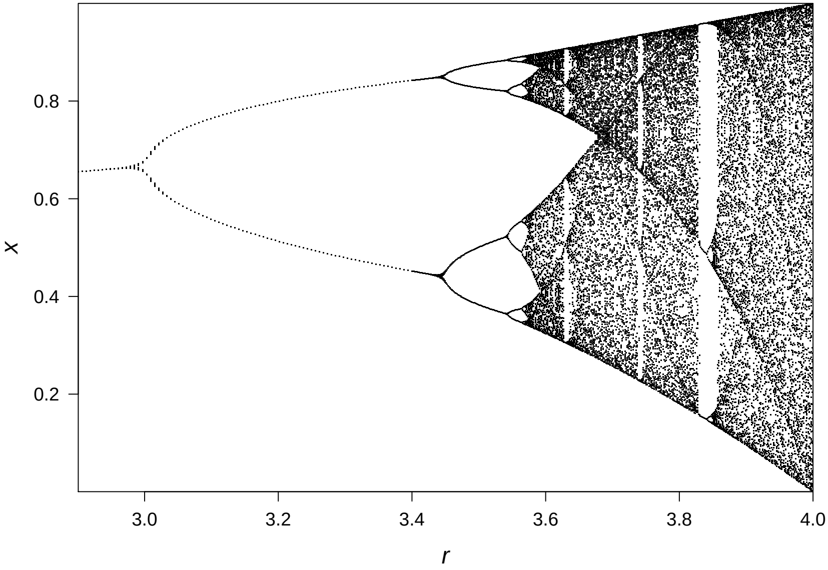
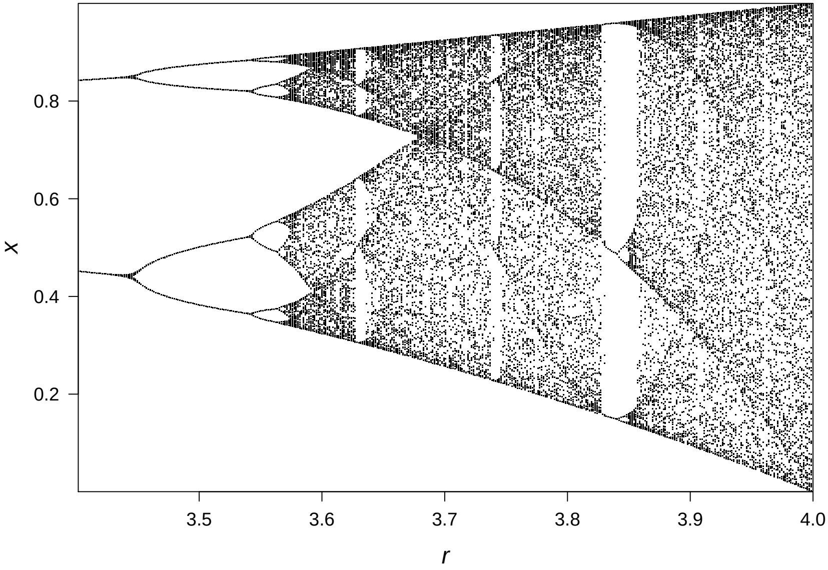


Logistic Map, $F(x) = rx(1-x)$, $\quad 2.9 \le r \le 4$



Logistic Map, $F(x) = rx(1-x)$, $3.4 \le r \le 4$



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- System is *coherent* (perfectly synchronous) if the state is the same in all patches
  i.e., $\mathbf{x}_1 = \mathbf{x}_2 = \cdots = \mathbf{x}_n$

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- *Colour coding of matrix indices:*
  - row indices are red
  - column indices are cyan

Basic properties of dispersal matrices $M = (m_{ij})$

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so any coherent state can be written $x\boldsymbol{e}$, for some $x \in \mathbb{R}$.

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Lemma (Row sums are the same)

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$\square$

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- Movie night?

Back to Space and Synchrony

Let's review what we've done so far on spatial models...

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  - If so, then this sort of analysis should help to identify synchronizing vaccination strategies.

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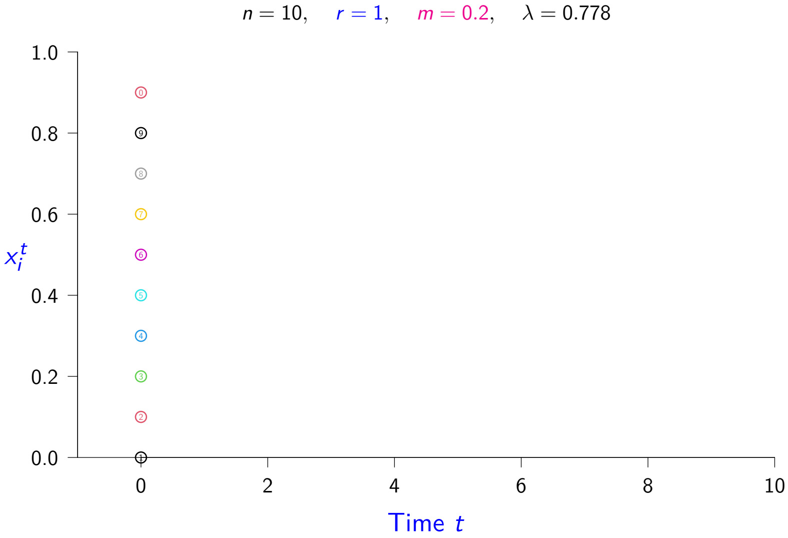

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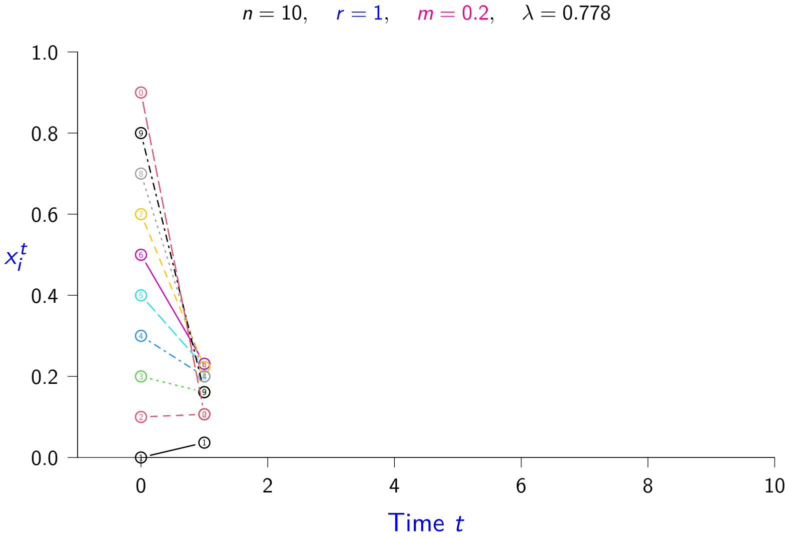

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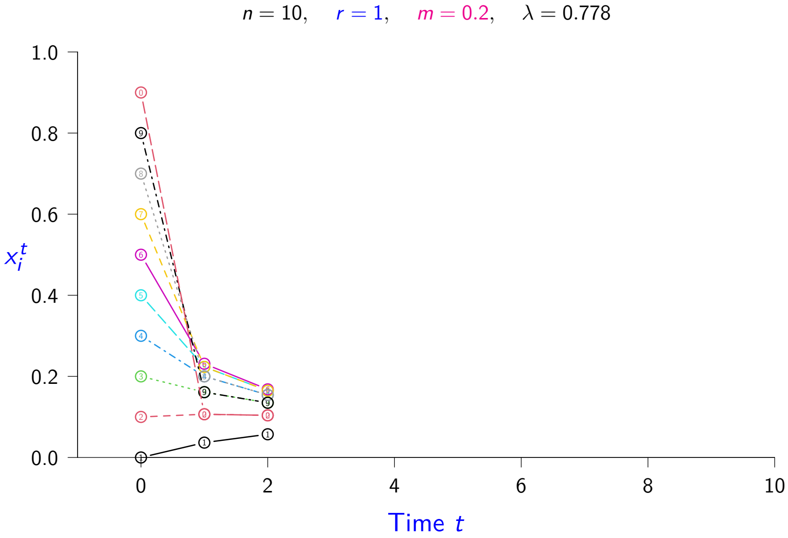

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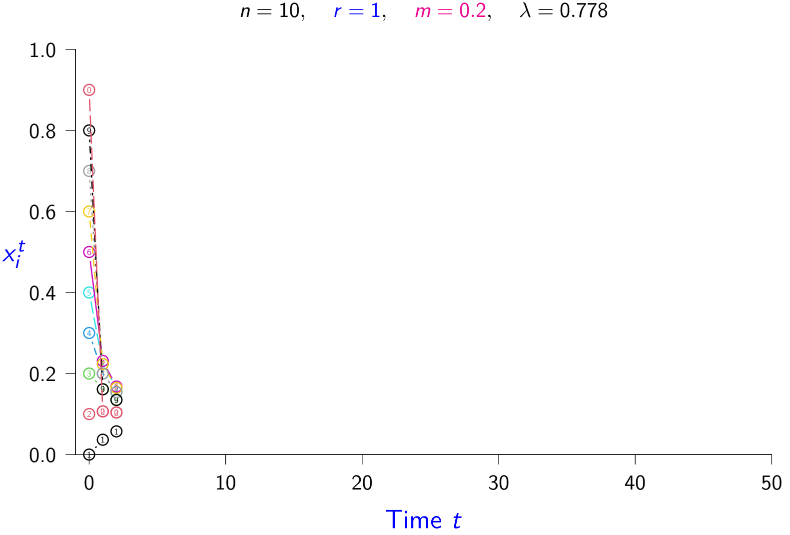

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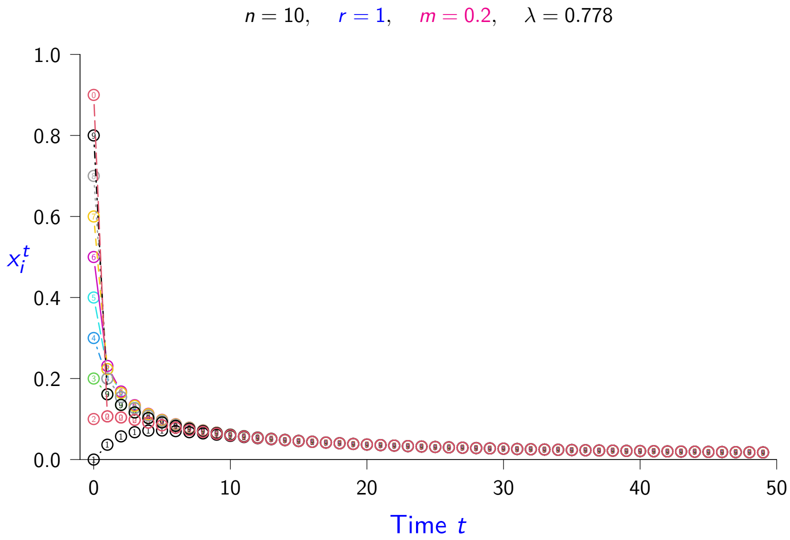

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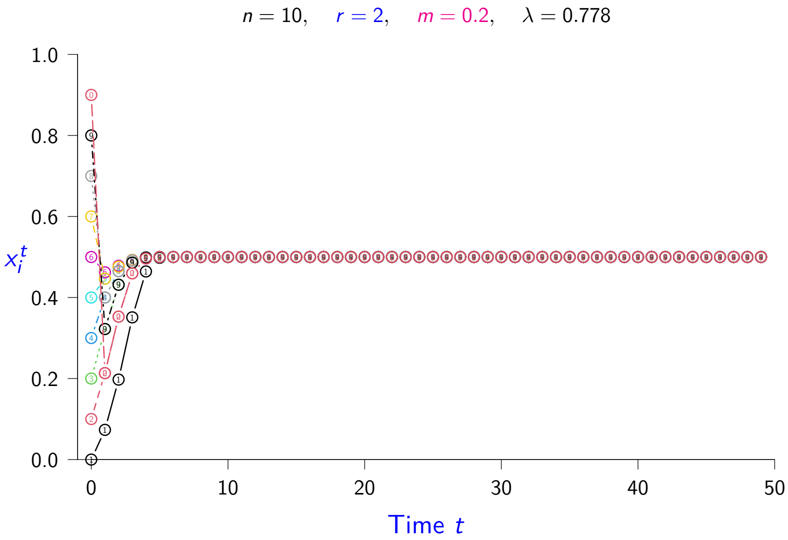

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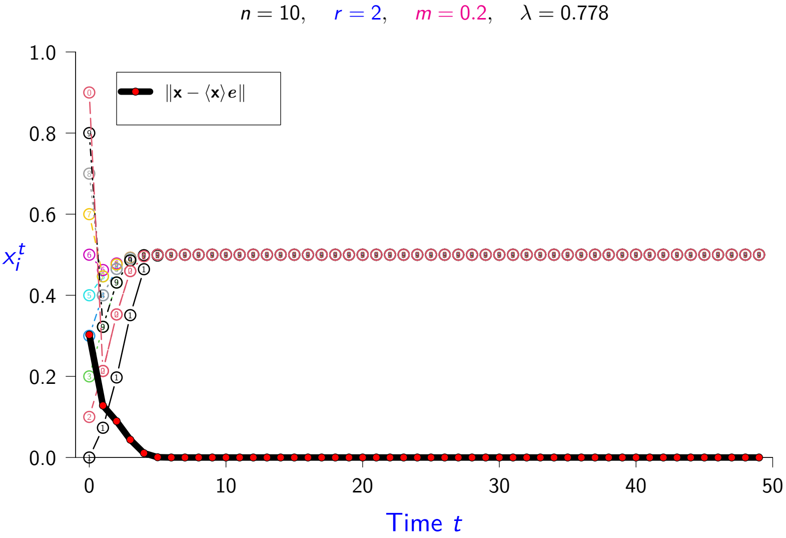

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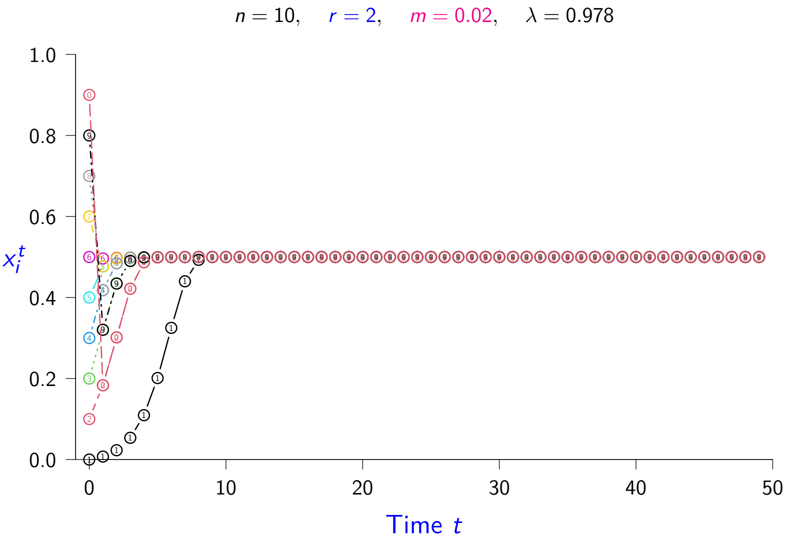

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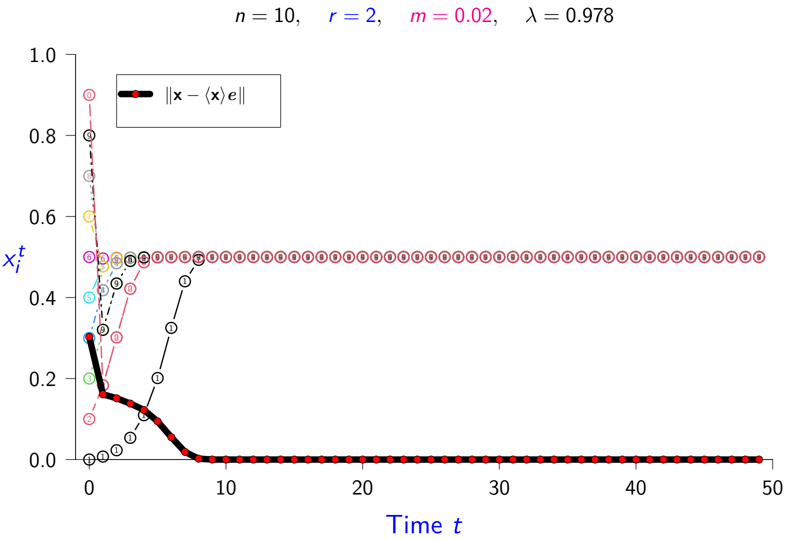

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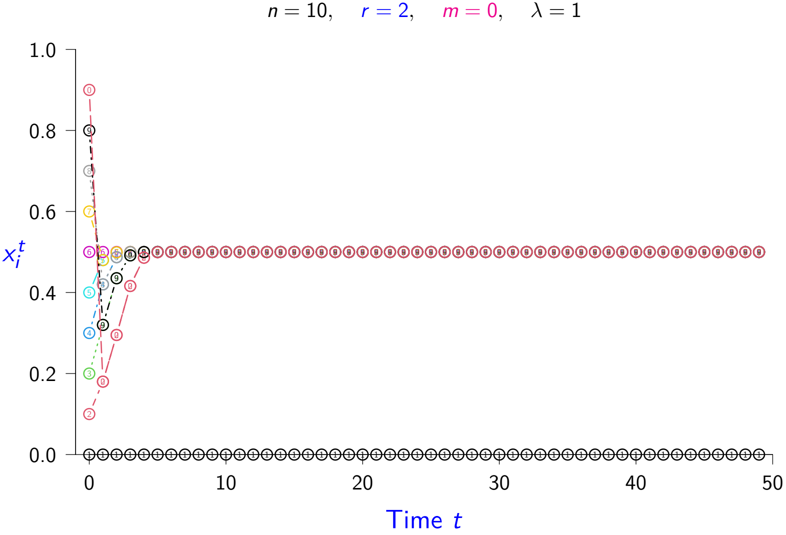

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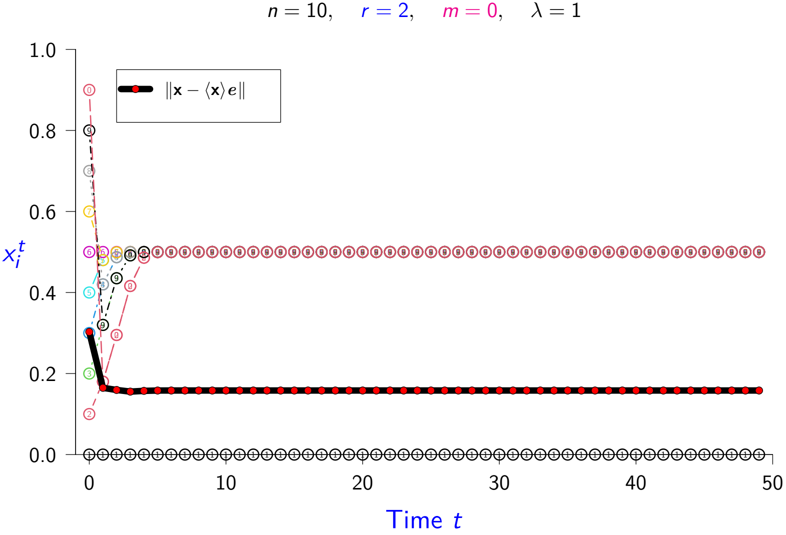

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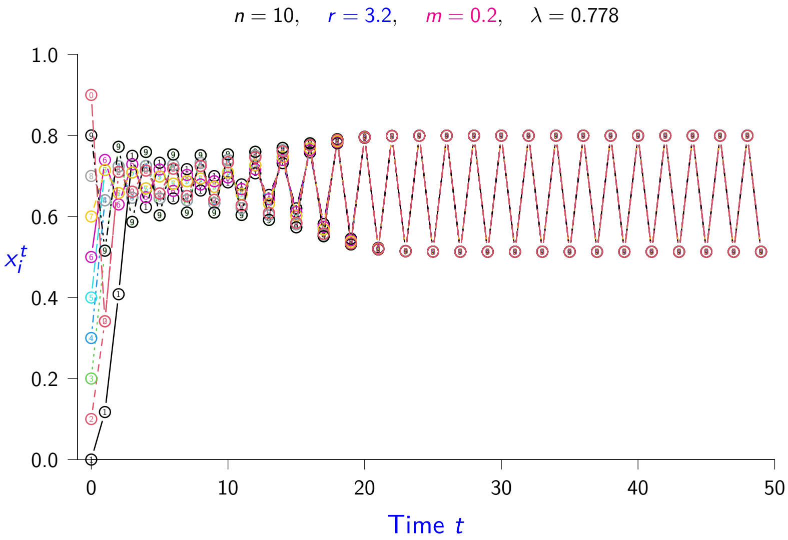

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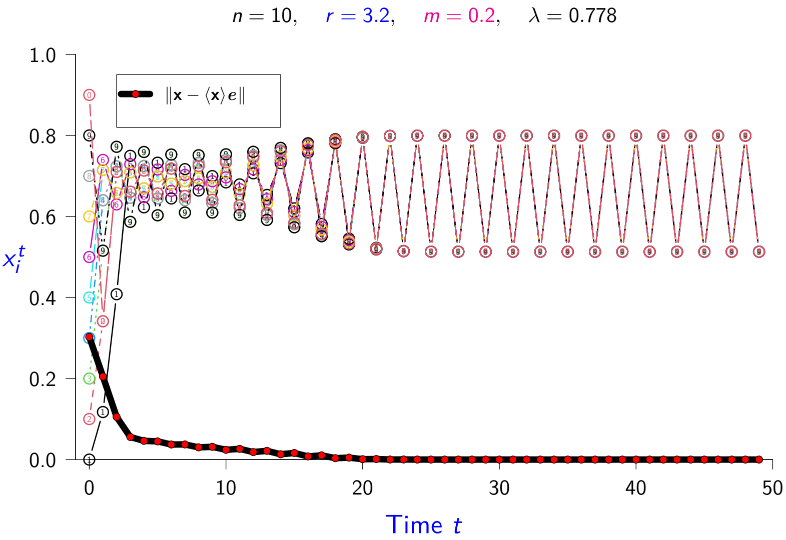

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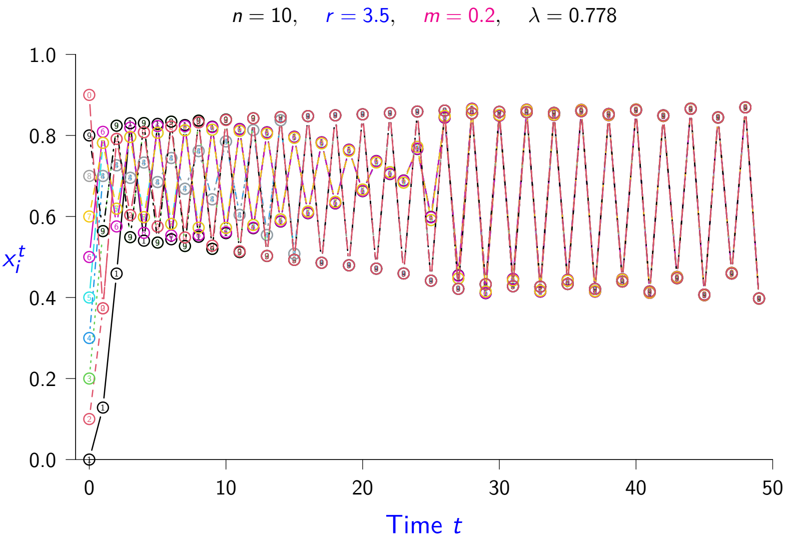

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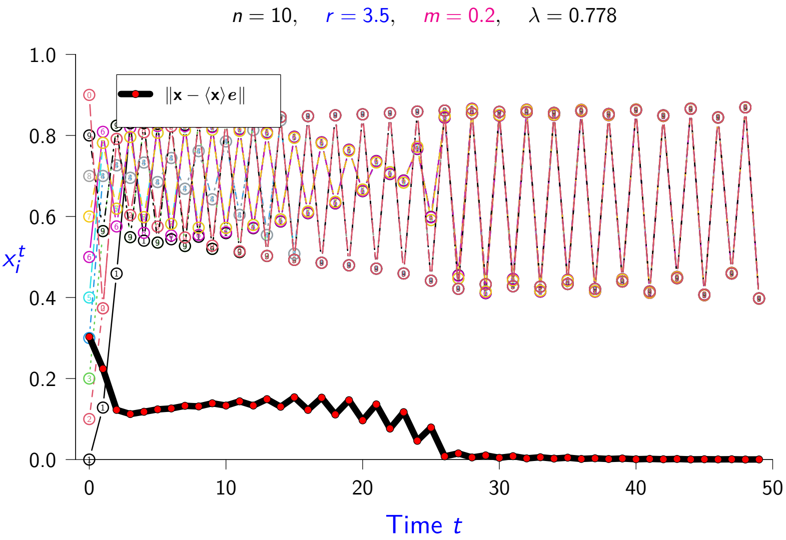

Logistic Metapopulation Simulation ($r = 3.75$, $m = 0.2$)

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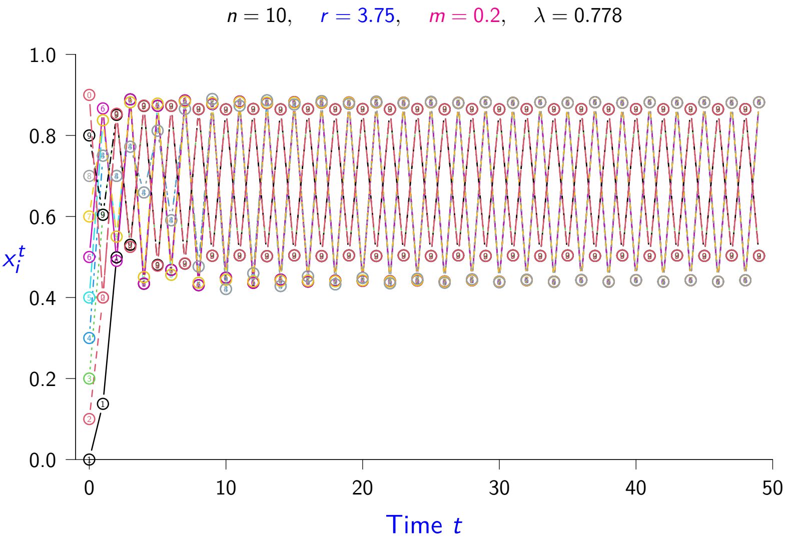

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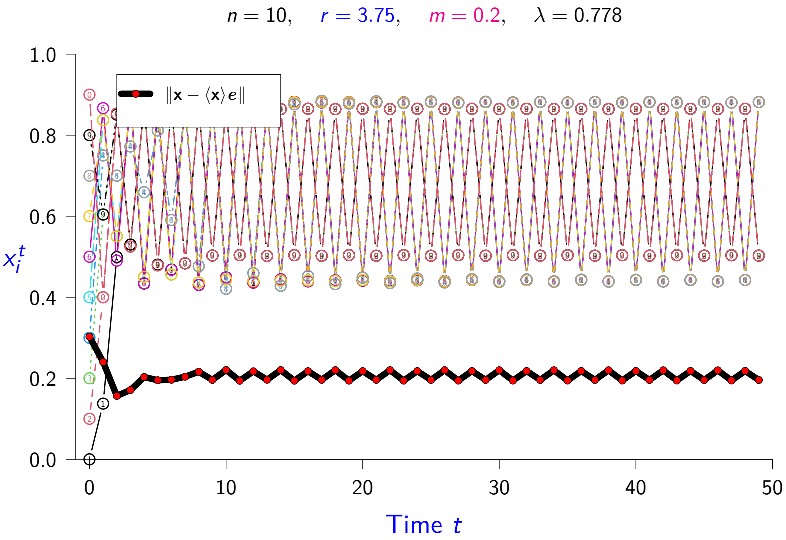

Logistic Metapopulation Simulation ($r = 3.83$, $m = 0.2$)

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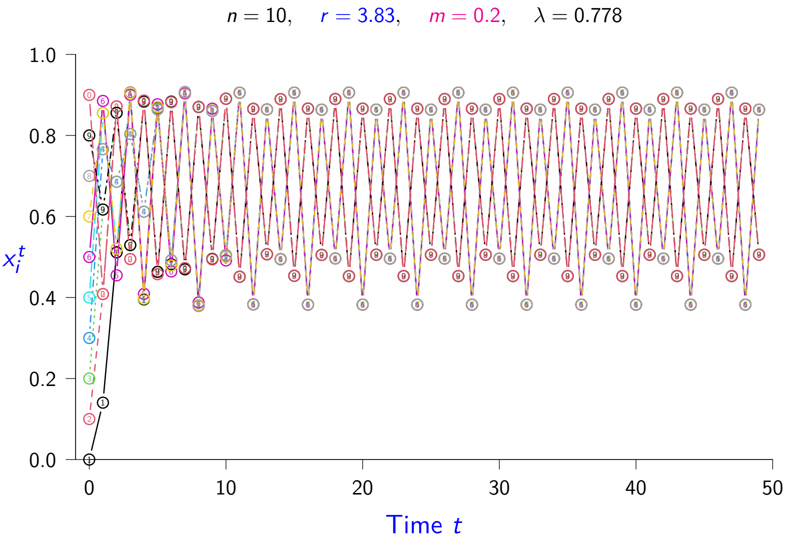

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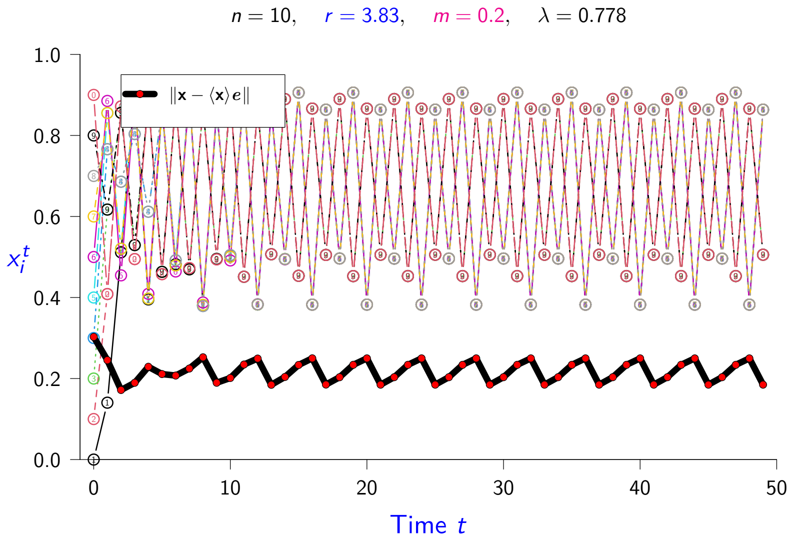

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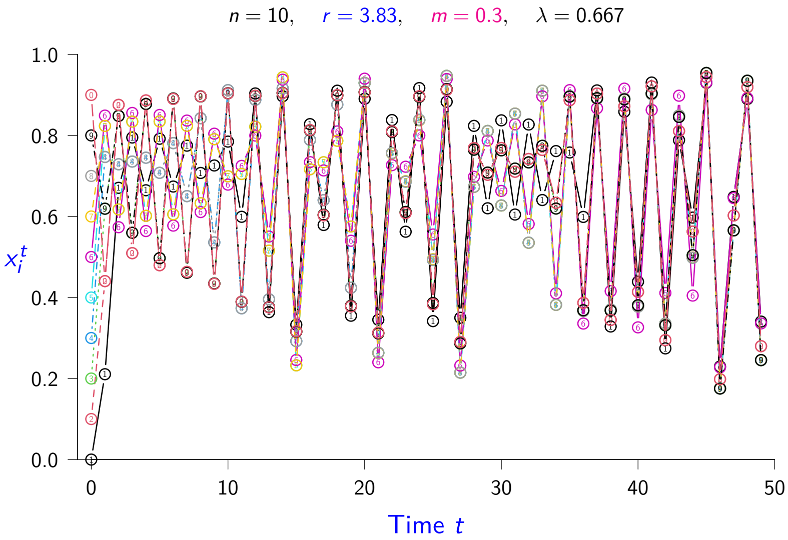

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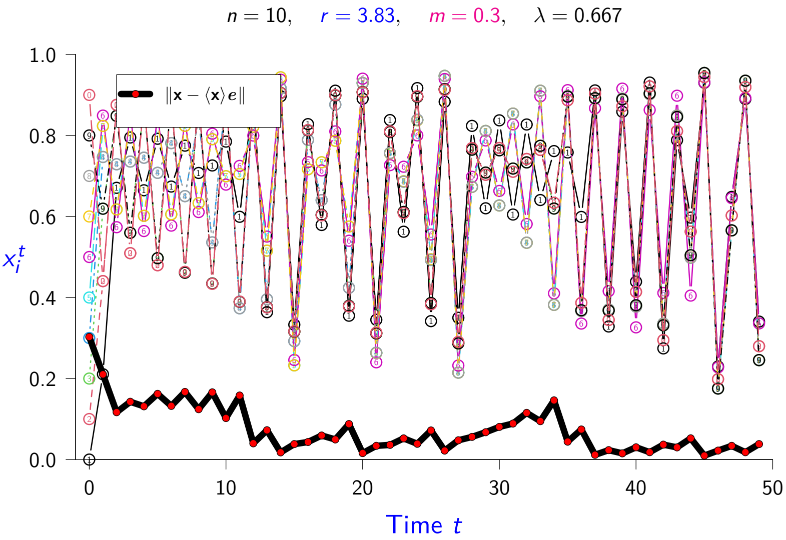

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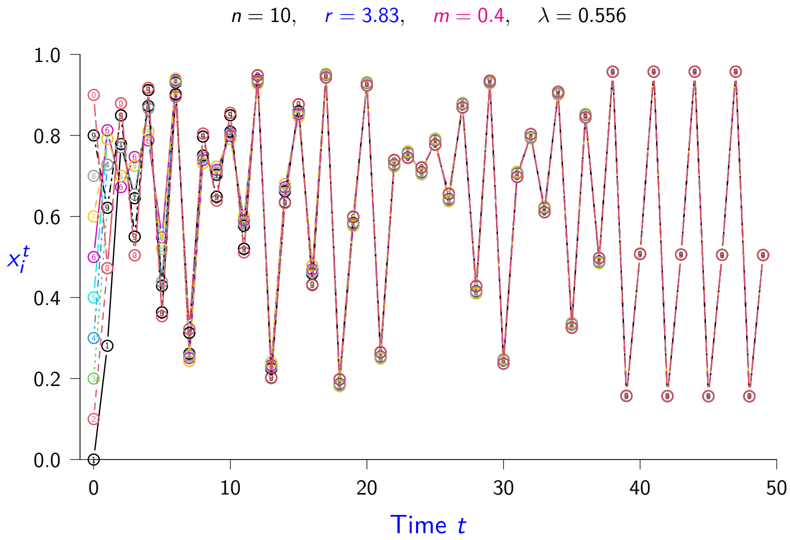

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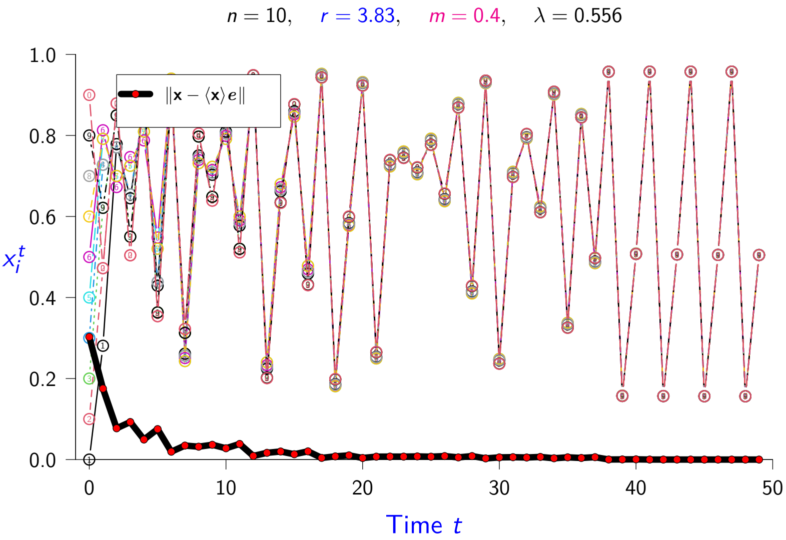

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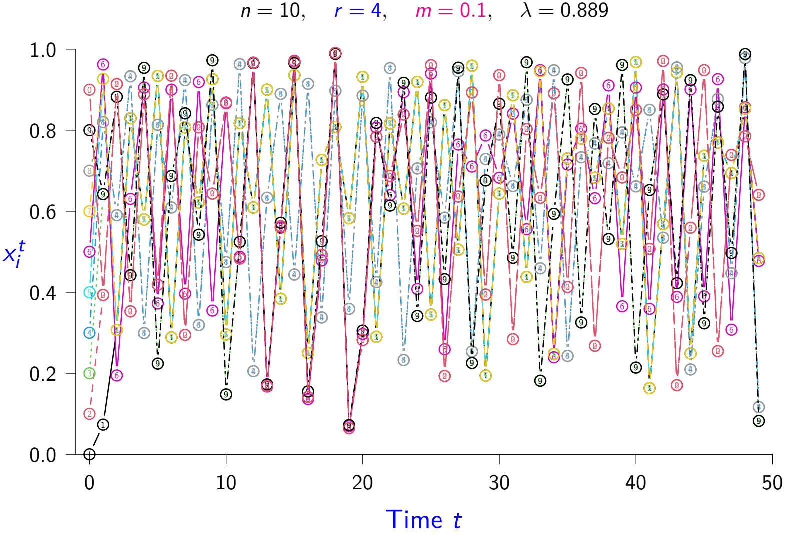

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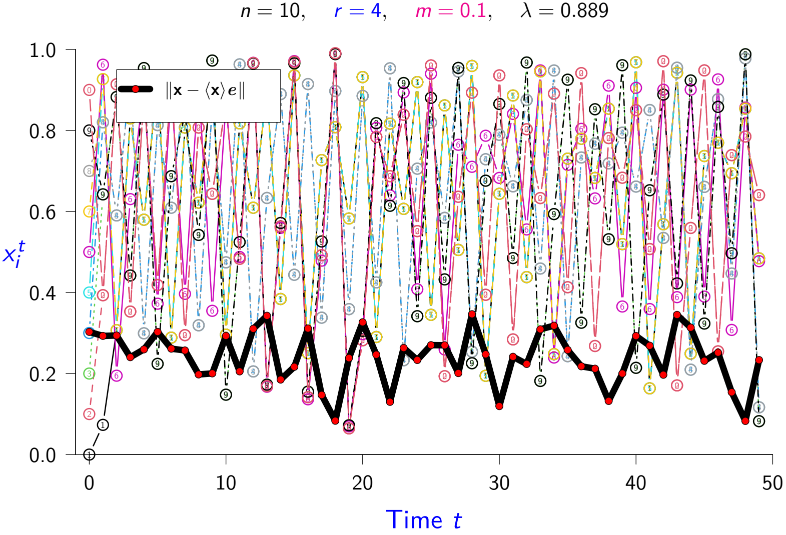

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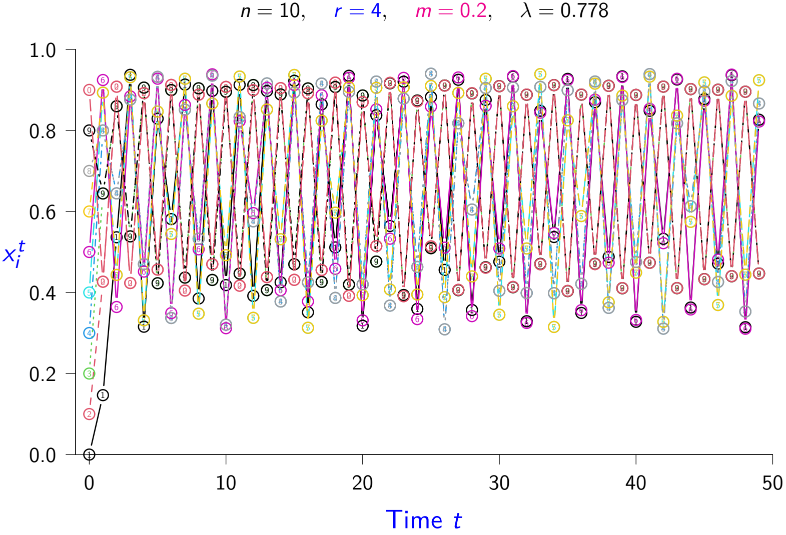

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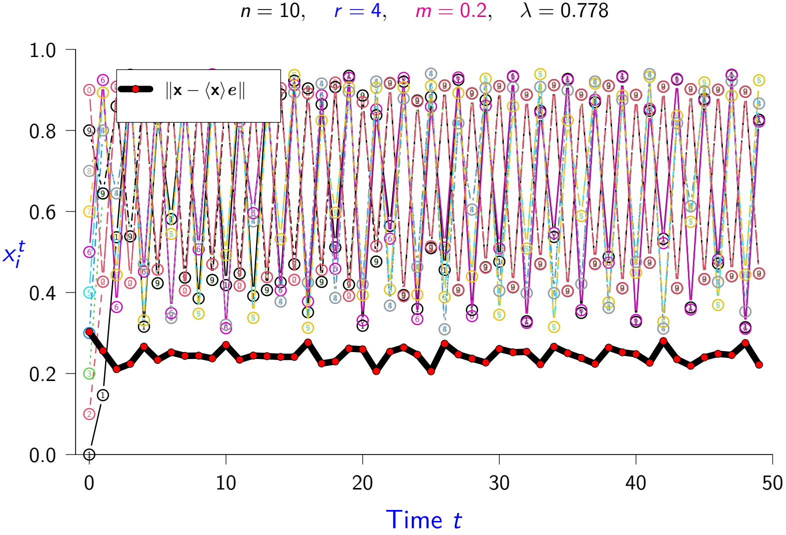

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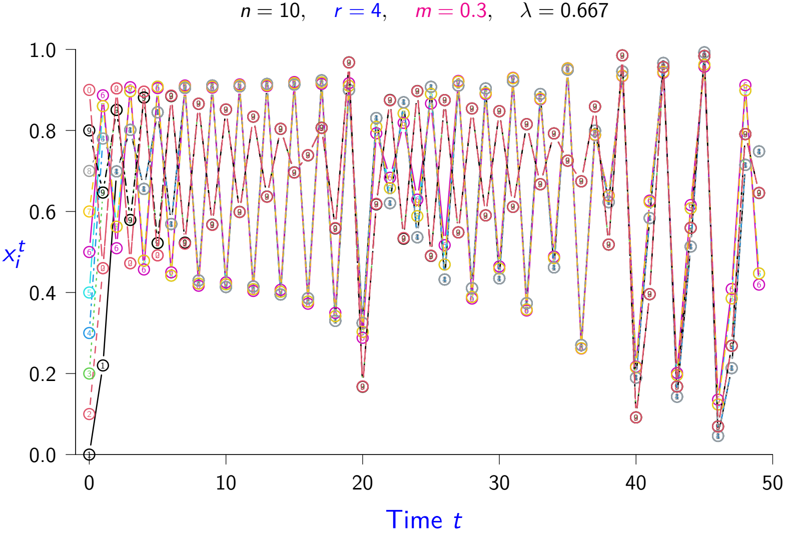

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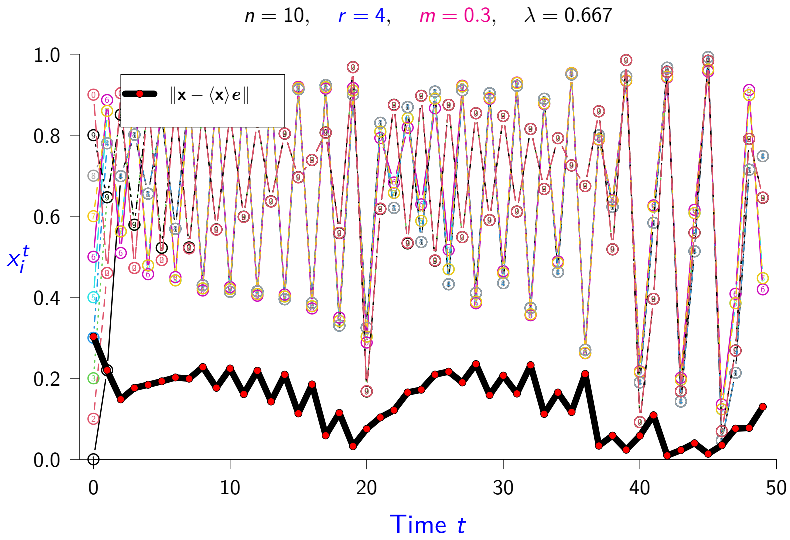

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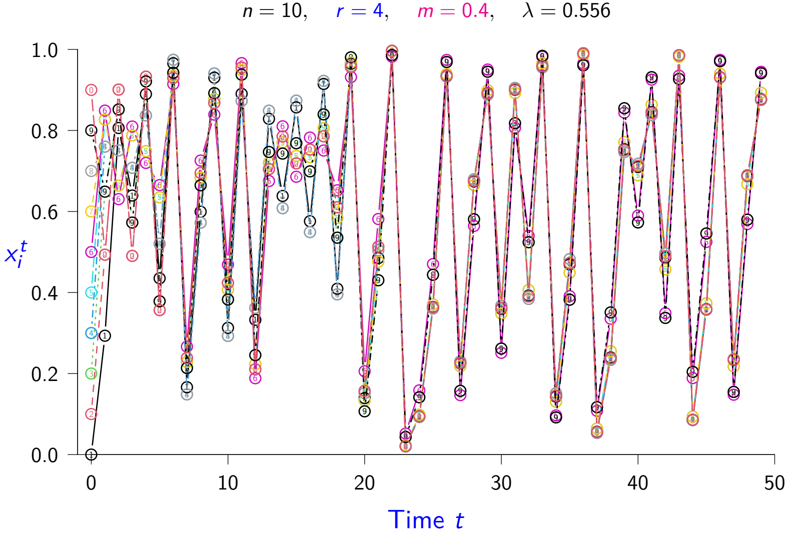

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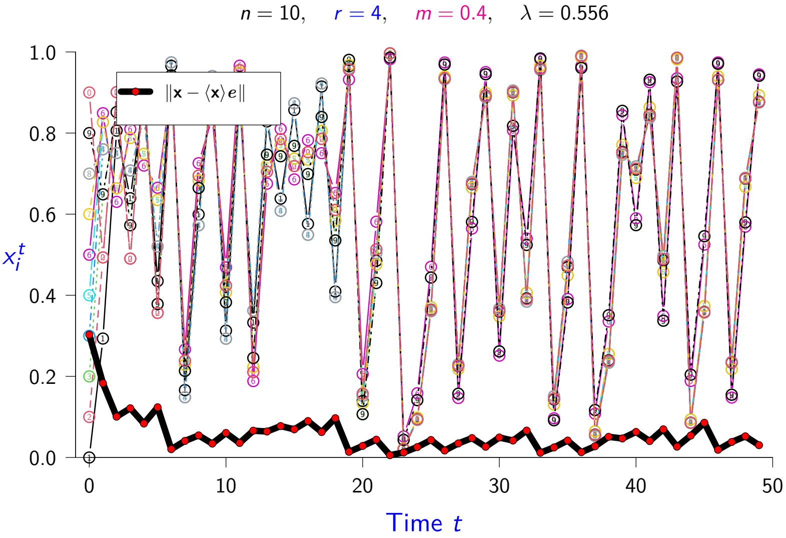

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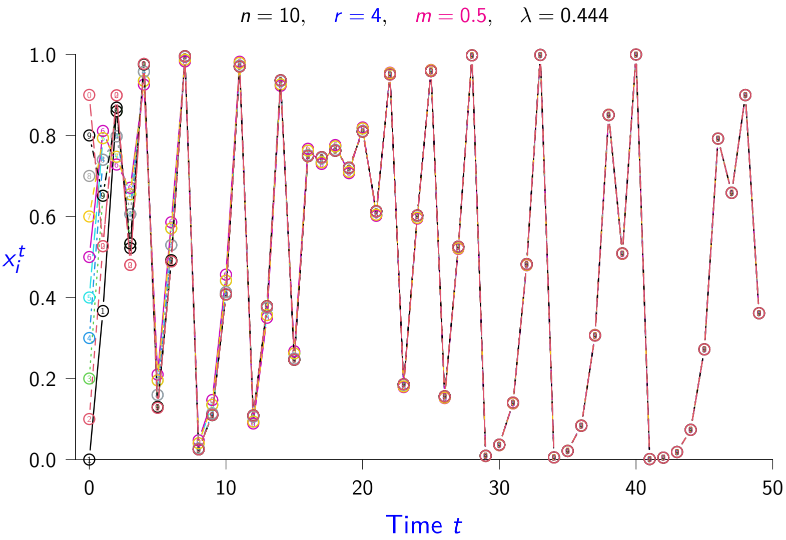

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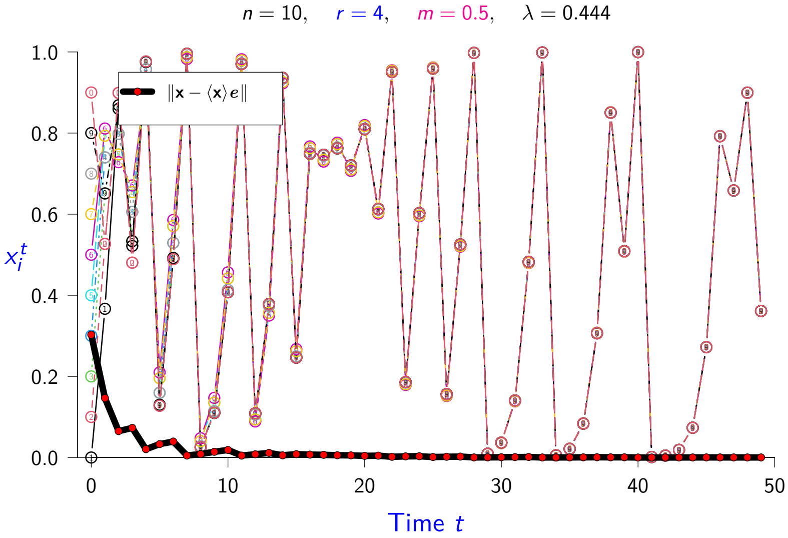

Metapopulation dynamics: what we've seen so far

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- Logistic Metapopulation Simulations (10 patches)

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- $r = 2, \ m = 0.2$
- $r = 2, \ m = 0.02$
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  - Next slide...
- Coherence is affected by magnitude $|\lambda|$ of *subdominant eigenvalue* λ.

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A vector is ***positive*** if each of its components is positive.

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Proof.

See Horn & Johnson (2013) *Matrix Analysis*, Corollary 8.1.30, p. 522. □

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Corollary

Consider a discrete-time metapopulation map,

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If solutions of the single patch system, $x^{t+1} = F(x^t)$, yield coherent solutions of $(\heartsuit)$ then 1 is a dominant eigenvalue of M.

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But $\boldsymbol{e}$ is a positive vector, hence by the lemma on the previous slide, 1 is a dominant eigenvalue of M. $\square$

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This is called the maximum (Lyapunov) *characteristic exponent* of the single patch map.

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Earn, Levin & Rohani (2000) *Science* **290**, 1360–1364

McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 9
Space II
Tuesday 5 November 2024

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  - Finding the initial growth rate for an epidemic model expressed with ODEs.
    - The initial growth rate r is the dominant eigenvalue of the linearization of this system at the DFE.
    - Simple to calculate if you've already computed $\mathcal{R}_0$ via the next generation matrix: as noted in class in Lecture 7 (final slide on estimating $\mathcal{R}_0$), r is the largest positive (or least negative) real part of the eigenvalues of $F - V$.

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  - Finding equilibria of discrete time models, *e.g.*, models of the form $x^{t+1} = F(x^t)$.

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  - Make sure you understand and can explain bifurcation diagrams with respect to seasonal amplitude (α) and with respect to basic reproduction number ($\mathcal{R}_0$). In particular, make sure you can explain how relevant bifurcation diagrams can be used to explain transitions in dynamics of infectious diseases that cause recurrent epidemics.

Midterm Test

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4mb-6mb-test-fb386

#1 Page 1 of 16

MATHEMATICS 4MB3/6MB3

Midterm Test, Tuesday 12 November 2024

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Special Instructions and Notes:

(i) This test has **16** pages. Verify that your copy is complete. Note that the final three pages are blank to provide additional space if needed.

(ii) **Answer all questions in the space provided.**

(iii) It is possible to obtain a total of 100 marks. There are 8 multiple choice questions (question 1 is worth 2 marks, whereas all other multiple choice questions are worth 4 marks each). There are 10 short answer questions (worth 7 marks each).

(iv) **For multiple choice questions, circle only one answer.**

(v) No calculators, notes, or aids of any kind are permitted.

(vi) PHAC refers to the Public Health Agency of Canada.

GOOD LUCK

Page 1 of 16

Coherence: what we've seen so far

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- Quantities that affect coherence

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- Quantities that affect coherence
- Coherence criteria

Global asymptotic coherence (GAC) for equal coupling

Theorem: $r|\lambda| < 1 \implies$ GAC.

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Proof in case of equal coupling:

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$$m_{ij} = \begin{cases} 1 - m & i = j \\ m/(n-1) & i \neq j \end{cases}$$

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Note: Actually true for very general connectivity matrices M and multi-dimensional single-patch dynamics $F(\mathbf{x})$.

Earn & Levin (2006) *PNAS* **103**, 3968-3971

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- More significant theoretically than practically, because it yields only *possibility* rather than *probability* of coherence.
- Quasi-global theory attempts to bridge the gap between "probability $= 1$" and "probability > 0".

McCluskey & Earn (2011) *J. Math. Biol.* **62**, 509–541

Application of simple coherence criteria

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10 patch logistic metapopulation

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$r|\lambda| < 1$ (coherence inevitable)

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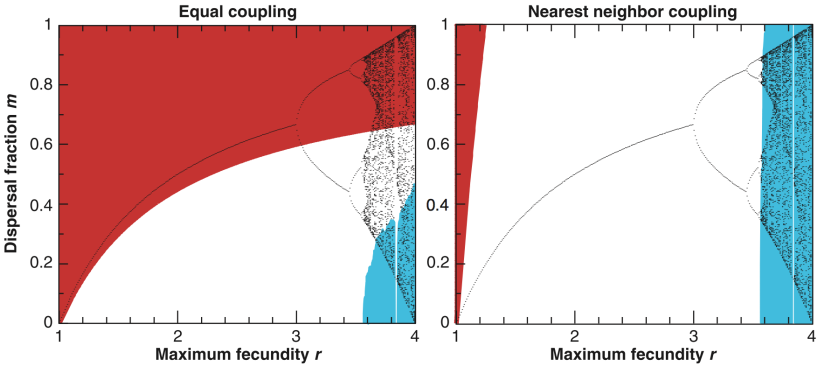

Earn, Levin & Rohani (2000) *Science* **290**, 1360–1364

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  McCluskey & Earn (2011) *J. Math. Biol.* **62**, 509–541

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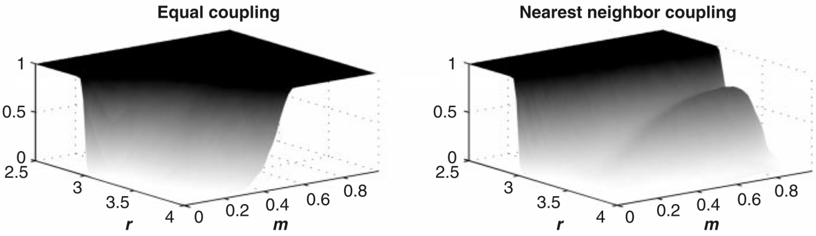
- Systematically explore representative set of initial conditions and determine probability of coherence within some tolerance, within some specified time
  - *e.g.*, coherence to within 10% within 10 iterations



Earn, Levin & Rohani (2000) *Science* **290**, 1360–1364

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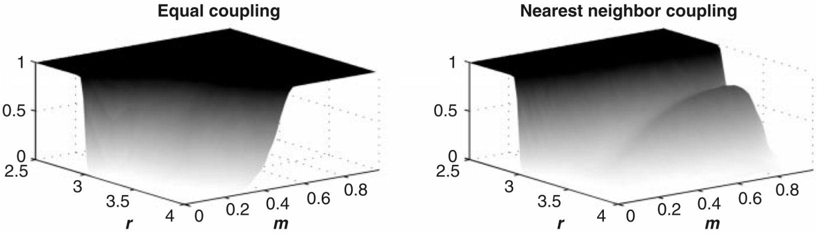

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- *Extremely demanding computationally. . .*

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- Global noise models *environmental stochasticity* (*e.g.*, weather), which presents a large risk of global extinction because the noise is correlated across all patches.

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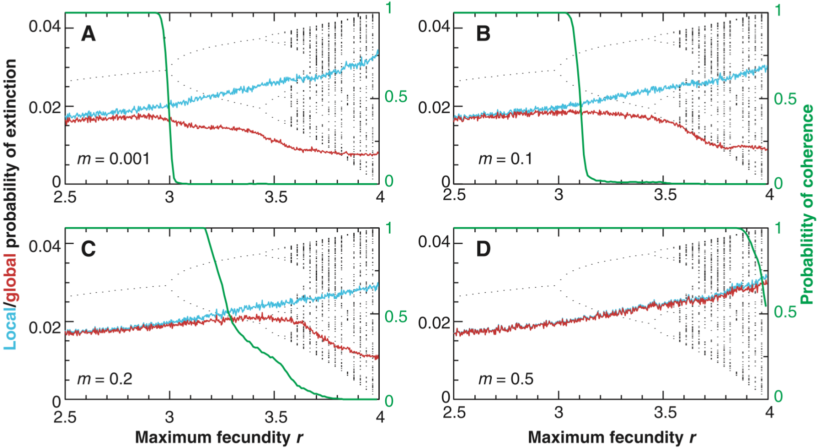

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Earn, Levin & Rohani (2000) "Coherence and Conservation" *Science* **290**, 1360–1364

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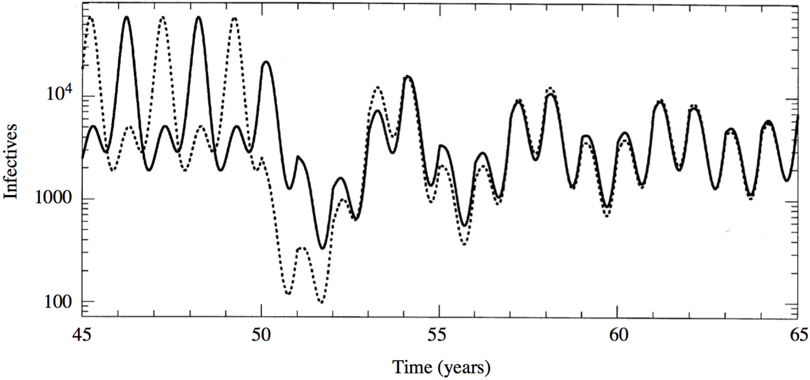
Example of Synchronization via Pulse Vaccination



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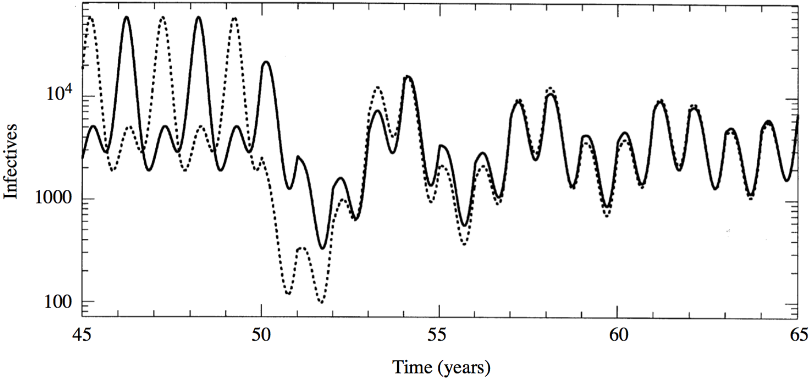

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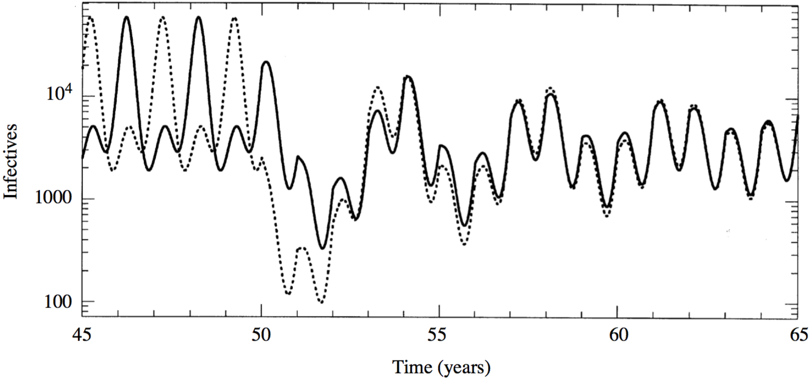

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